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DETECTION OF GRAVITATIONAL WAVES WITH THE MOON AS AN ELASTIC BODY

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Abstract

In this thesis, we review in detail the interaction between gravitational waves (GWs) and the Moon, treated as an elastic body, as presented in a paper by M. Kachelriess and M. P. Nødtvedt [1]. The aim is, to combine the content of this paper with additional findings from previous literature to make the topic more accessible.
Building on the theoretical framework of elasticity, we derive and linearize Cauchy's equations of motion to model the Moon's response to such a GW. By investigating toroidal and spheroidal oscillations induced by a GW in different polarizations, we derive the differential equation systems to predict the Moon's displacements, which can then be numerically solved. The results of different papers are summarized to get the eigenfrequencies and modes of these displacements. We furthermore provide some insights on frequency-dependent response.

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1

Introduction

Albert Einstein predicted gravitational waves in 1916 [2], a year after the general relativity field equations [3]

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \tag{1.1}$$

were formulated. In linearized weak-field gravity, where we take [4]

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad |h_{\mu\nu}| \ll 1, \tag{1.2}$$

the field equations have solutions in the form of waves that travel at the speed of light due to, for example, variations in the source's mass quadrupole moment over time. In vacuum ($T_{\mu\nu} = 0$), we get

$$\Box h_{\mu\nu} = 0. \tag{1.3}$$

General relativity predicts that gravitational waves (GWs) propagating in $\hat{z}$-direction only have two polarizations: tensor plus and cross modes ($A_+, A_\times$) [5]. General gravity theories, however, can include up to four additional polarizations: three vector modes and one scalar mode (A_S). Since scalar extensions of Einstein gravity are much more popular than vector extensions, we will simplify our following discussions by neglecting the vector components. The polarization tensor has then the form [6, 7]:

$$\mathcal{E}_{ij} = \begin{pmatrix} A_S + A_+ & A_\times & 0 \\ A_\times & A_S - A_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{1.4}$$

Gravitational waves are then described by a plane wave

$$h_{ij} = \mathrm{Re}\{h_0 \mathcal{E}_{ij} e^{-i\vec{k}_0 \cdot \vec{r}}\}. \tag{1.5}$$

Advancements in technology, notably the construction of detectors like LIGO and Virgo [8], allowed us to observe gravitational waves directly.
In this thesis, however, we investigate another type of gravitational wave detection: the interaction between gravitational waves and elastic bodies. We aim to understand how these waves affect the behavior of spherical elastic bodies, in particular the Moon.

2

Theory of Elastic Bodies

This chapter introduces the theoretical framework for the response and deformation of spherical elastic bodies (which is a good model for the Moon, for example) and their response to gravitational waves. We derive Cauchy's equation of motion, which describes such displacements of elastic bodies under stresses and forces. We then linearize this equation and introduce methods for solving it, along with appropriate boundary conditions.

2.1 Stress and Strain Tensor

In the study of elasticity, two fundamental concepts are the stress tensor and the strain tensor. These tensors describe the internal forces and deformations in a material, respectively.

Stress Tensor

The stress tensor σ_{ij} is a second-order tensor that represents the internal forces within a deformable body. Each component σ_{ij} represents the force per unit area acting on the face perpendicular to the i-th coordinate axis in the j-th direction. The stress tensor is symmetric, i.e., $\sigma_{ij} = \sigma_{ji}$ and in three dimensions it is given by

$$\sigma = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}. \tag{2.1}$$

Strain Tensor

The strain tensor ε_{ij} is another second-order tensor that quantifies the deformation of a material. Each component ε_{ij} measures the relative infinitesimal

displacement of the material in the i-th direction relative to the j-th direction. Mathematically, the strain tensor is defined as [9]:

$$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i}). \tag{2.2}$$

where u_i is the displacement vector[1]. The relationship between the stress tensor and the strain tensor in a linear elastic material is given by Hooke's law in three dimensions which can be written as

$$\sigma_{ij} = \lambda\, \delta_{ij}\, \varepsilon_{kk} + 2\mu\, \varepsilon_{ij} \tag{2.3}$$

where λ and μ are the so-called Lamé parameters. With 2.2 we get the stress strain relation for an isotropic material

$$\sigma_{ij} = \lambda\, \delta_{ij}\, u_{k,k} + \mu(u_{i,j} + u_{j,i}). \tag{2.4}$$

The Lamé parameters are material dependent and are determined for each material from experiments. The stress tensor can be then fully determined from the displacement u_i and the material properties described in the Lamé parameters.

2.2 Gravitational Waves and Elastic Bodies

Gravitational waves, predicted by general relativity, interact with matter by inducing tidal forces. This interaction causes distortions in the shape of massive objects, including elastic bodies. In the context of non-relativistic elastic isotropic bodies, gravitational waves perturb the material by exerting forces that induce displacements in the body's constituent elements.
Consider a spherical elastic body with density ρ and initial radius R_0. Gravitational waves propagating through the body induce strains and stresses, causing changes in the body's shape and size. The displacement of a body element from its equilibrium position due to the gravitational wave can be described by the displacement vector u_i.
The Lagrangian density $\mathcal{L}$ describing the dynamics of the elastic body is given by [10]

$$\mathcal{L}_{\mathrm{GW}} = \frac{1}{2}\rho \dot{u}_i \dot{u}^i - \frac{1}{2}\varepsilon_{ij}\sigma^{ij}, \tag{2.5}$$

[1] If $r_i^* = f(r_i)$ is a deformation of a point $r_i \in \Omega$, where Ω is the body of interest, then the displacement vector is given by $u_i = r_i^* - r_i$.

where $\dot{u}_i$ represents the time derivative of the displacement vector u_i, and σ^{ij} is the stress tensor induced by the gravitational wave. The Lagrangian of an elastic body in an external gravitational perturbation $h_{\mu\nu}$ can then be obtained by using the post-Newtonian approximation, where we assume that the GW acts as an additional strain on the body. Therefore ε_{ij} is replaced by $\varepsilon_{ij} + h_{ij}$ obtaining [10]

$$\mathcal{L}_{\text{pert}} = \frac{1}{2}\rho\dot{u}_i\dot{u}^i - \frac{1}{2}(\varepsilon_{ij} + h_{ij})\sigma^{ij}. \tag{2.6}$$

We get the equation of motion

$$\partial_t(\rho\dot{u}^i) = \nabla_j\sigma^{ij} - \nabla_j(\mu h^{ij}) - \frac{1}{2}\nabla^i(\lambda h), \tag{2.7}$$

assuming $h_{\mu 0} = h_{0\mu} = 0$. All of this calculation was done, without using the TT gauge for $h_{\mu\nu}$ to allow theories of modified gravity. The force density due to the gravitational perturbation is given by the last two terms

$$f_i = -\nabla_j(\mu h_i^j) - \frac{1}{2}\nabla^i(\lambda h) \tag{2.8}$$

with corresponding stress

$$\sigma_{ij} = -\mu h_{ij} - \frac{1}{2}\lambda h\delta_{ij}. \tag{2.9}$$

In Einstein gravity, the trace of $h_{\mu\nu}$ vanishes ($h = 0$), so the last terms in both equations vanish as well.

2.3 Euler's and Cauchy's Equation of Motion

Consider a continuous distribution of matter in motion within an inertial reference frame. Let S be the boundary enclosing volume V. Furthermore, let the density field be $\rho(\mathbf{r}, t)$ and the velocity field $v_i(\mathbf{r}, t)$. We focus on the total change of a fluid field A along a trajectory over time interval dt. For a moving reference frame, we define the covariant derivative D/Dt as

$$\frac{DA}{Dt} := \frac{\partial A}{\partial t} + v_i\frac{\partial A}{\partial x_j}. \tag{2.10}$$

This is essentially the derivative that describes the change of the fluid while being in the moving reference system of the fluid A. The total mass m, linear momentum $P_i(t)$, and angular momentum $M_k(t)$ are given by

$$m = \int_V \rho dV, \quad P_i(t) = \int_V v_i\rho dV, \quad M_k(t) = \int_V (\epsilon_{ijk}r_iv_j)\rho dV. \tag{2.11}$$

Assuming mass conservation, total force Λ, and total torque Γ, the following hold within V:

$$\begin{aligned} \frac{Dm}{Dt} &= 0, \\ \frac{DP_i(t)}{Dt} &= \Lambda_i, \\ \frac{DM_i(t)}{Dt} &= \Gamma_i. \end{aligned} \tag{2.12}$$

The force Λ_i includes a combination of a body force term ρF_i and a surface force $n_i \sigma^{ij}$ acting on the boundary S, so

$$\Lambda_i = \int_S (\sigma_{ij} dS_i) + \int_V \rho F_i dV = \int_V (\sigma_{ij,j} + \rho F_i) dV, \tag{2.13}$$

where we used Gauss's theorem in the last equality. Similarly, the torque Γ_k can be expressed as

$$\Gamma_k = \int_S \epsilon_{ijk} r_i (dS_a \sigma_{aj}) + \int_V \epsilon_{ijk} r_i \rho F_j dV. \tag{2.14}$$

Applying the covariant derivative to mass conservation 2.12 gives:

$$\frac{Dm}{dt} = \frac{D}{Dt} \int_V \rho dV = \int_V \left[\frac{D\rho}{Dt} + \rho v_{i,i} \right] dV = \int_V \left[\frac{\partial \rho}{\partial t} + (\rho v_i)_{,i} \right] dV = 0. \tag{2.15}$$

Let us now consider the covariant derivative of $P_i(t)$. Using again 2.12 and the second equality of 2.15 we get

$$\begin{aligned} \Lambda_i &= \frac{D}{Dt} P_i(t) = \frac{D}{Dt} \int_V v_i \rho dV = \int_V \left[\frac{D(\rho v_i)}{Dt} + \rho v_i v_{j,j} \right] dV \\ &= \int_V \left[\rho \frac{Dv_i}{Dt} + v_i \left(\frac{D\rho}{Dt} + \rho v_{j,j} \right) \right] dV = \int_V \left(\rho \frac{Dv_i}{Dt} \right) dV. \end{aligned} \tag{2.16}$$

With 2.13 we obtain Euler's equation of motion:

$$\sigma_{ij,j} + \rho F_i = \rho \frac{Dv_i}{Dt}. \tag{2.17}$$

We can still make some assumptions for this paper. First, we will assume $v_i(\mathbf{r}, t) = \frac{Du_i}{Dt} \approx \frac{\partial u_i}{\partial t}$, which is valid when $\left| \frac{\partial u_i}{\partial x_j} \right| \ll 1$. This leads to

$$\frac{Dv_i}{Dt} \approx \frac{\partial^2 u_i}{\partial t^2}. \tag{2.18}$$

Using the Fourier transform for displacement $u_i(t)$ gives the equation:

$$\sigma_{ij,j} + \rho F_i + \rho\omega^2 u_i = 0. \tag{2.19}$$

This is Cauchy's equation of motion for the infinitesimal theory of elasticity, used to describe for example Moon oscillations, assuming spherical shapes and a spherical coordinate system (r, θ, ϕ), with density ρ and Lamé parameters λ and μ.

2.4 Modes and Hansen Vectors

The displacements of a spherically symmetric body are characterized by a set of eigenfunctions, $u^{(nml)}(\mathbf{r}, t)$, which are specified by the three numbers nml. The displacement vector $u_i(\mathbf{r}, t)$ can then be written as

$$u_i(\mathbf{r}, t) = \sum_{k=nml} u^{(k)}(\mathbf{r}, t). \tag{2.20}$$

The time-dependence of small oscillations, i.e.

$$u_i^{(nml)}(\mathbf{r}, t) = u_i^{(nml)}(\mathbf{r})\bar{g}(t) \tag{2.21}$$

can be modeled as a damped harmonic oscillator with eigenfrequency ω_n. This will be further discussed in chapter 3. We will for now try to find a better way to write these eigenfunctions in terms of vector surface harmonics in spherical coordinates. The displacement $u_i^{(nml)}$ on a homogenous isotropic elastic solid subjected to the body force distribution $F^i(\mathbf{r}, t)$ can be described by the Navier-Lamé equation, which in Fourier space is given by (see equation 2.19 and 2.4 for comparison, this is an even more general result)

$$\alpha^2(u_{i,i}^{(nml)})_{,j} - \beta^2\epsilon_{jak}(\epsilon_{abc}u_{c,b}^{(nml)})_{,k} + F_j + \omega^2 u_j^{(nml)} = 0, \tag{2.22}$$

where α respectively β are dependent on the Lamé parameters λ and μ and the density of the body. Hansen (1935) proposed a method for generating solutions of such vector wave equations in any co-ordinate system [11, 12]. The three Hansen vectors M_i, N_i and L_i are defined as

$$\begin{aligned} M_i &= \epsilon_{ibc}\psi a_{c,b} \\ N_i &= \frac{1}{k^{(c)}}\epsilon_{iak}(\epsilon_{abc}\psi a_{c,b})_{,k} \qquad k^{(c)} = \frac{\omega}{c} \\ L_i &= \frac{1}{k^{(c)}}\psi_{,i}, \end{aligned} \tag{2.23}$$

where the potential ψ is a solution to the scalar Helmholtz equation

$$\psi_{,jj} + k^{(c)}\psi = 0 \tag{2.24}$$

and $a_i = r\hat{e}_i^{(r)}$ in spherical coordinates. We can separate the potential into a radial and angular part $\psi = f(r)Y(\theta,\phi)$, obtaining [13]

$$\begin{aligned} M_i &= fC_i \\ k^{(c)}N_i &= \left[(k^{(c)})^2(rf) + \frac{\partial^2}{\partial(rf)r^2}\right]P_i + \frac{1}{r}\frac{\partial(fr)}{\partial r}B_i, \\ k^{(c)}L_i &= \left(\frac{\partial f}{\partial r}\right)P_i + \frac{1}{r}fB_i, \end{aligned} \tag{2.25}$$

where C_i, P_i and B_i are the vector surface harmonics

$$\begin{aligned} \sqrt{l(l+1)}C_i^{(ml)}(\theta,\phi) &= \epsilon_{ijk}(r_{k,j}Y(\theta,\phi)), \\ P_i^{(ml)}(\theta,\phi) &= \hat{e}_i^{(r)}Y^{(ml)}(\theta,\phi), \\ \sqrt{l(l+1)}B_i^{(ml)}(\theta,\phi) &= rY_{,i}^{(ml)}(\theta,\phi). \end{aligned} \tag{2.26}$$

The Hansen vectors are linearly independent and a general displacement can be written as a linear combination of them

$$u_i^{(nml)}(\mathbf{r},t) = \sum_{\sigma,m,l} \alpha^{(ml)}M_i^{(ml)} + \beta^{(ml)}N_i^{(ml)} + \gamma^{(ml)}L_i^{(ml)} \tag{2.27}$$

where α, β and γ are some constants to be determined. Alternatively, since Hansen vectors depend entirely on the vector surface harmonics, we can also write our general solution as a linear combination of them

$$\begin{aligned} u_i^{(nml)}(\mathbf{r}) = U^{(n)}(r)P_i^{(ml)}(\theta,\phi) \\ + V^{(n)}(r)\sqrt{l(l+1)}B_i^{(ml)}(\theta,\phi) \\ + W^{(n)}(r)\sqrt{l(l+1)}C_i^{(ml)}(\theta,\phi). \end{aligned} \tag{2.28}$$

By inserting this general solution into the Navier-Lamé equation we can get equations for $U(r)$, $V(r)$ and $W(r)$. Oscillations, in which the displacement is only in the form of $V(r) = W(r) = 0$, are called toroidal. If we have $U(r) = 0$ instead, the oscillation is spheroidal. The Hansen vectors take on specific forms based on the system's symmetries. We will use the form that preserves the spherical symmetry of the Moon.

2.5 Linearization of Cauchy's equation of motion

To simplify equation 2.19 we will make some assumptions for our spherical body. We denote unperturbed quantities with the subscript zero:

1. We neglect small quantities of second or higher order.
2. The initial stress $\sigma_{jk}^{(0)}$ is in equilibrium and described by hydrostatic pressure:

$$\sigma_{jk}^{(0)} = -p^{(0)}\delta_{jk}, \quad \text{with} \quad \partial_j p^{(0)} = -g^{(0)}\rho^{(0)}\hat{e}_j^{(r)}. \tag{2.29}$$

3. A small element maintains its initial stress during displacement and may get additional stresses due to compression and distortion, thus $p^{(0)}(r) = p^{(0)}(r-u)$.
4. The total stress in a deformed state can be split into initial equilibrium stress and an additional elastic stress. The initial stress is

$$\sigma_{jk}^{(0)}(r) = -p^{(0)}(r-u)\delta_{jk}, \tag{2.30}$$

   by using the second and third assumptions, and the additional elastic stress follows equation 2.4:

$$\delta\sigma_{jk}(u) = \lambda\delta_{jk}u_{i,i} + \mu(u_{j,k} + u_{k,j}). \tag{2.31}$$

5. Gravity is the only body force, derived from the gravitational potential Ψ:

$$F_j = \partial_j\Psi. \tag{2.32}$$

6. The density ρ and gravitational potential Ψ are split into equilibrium components ($\rho^{(0)}$ respectively $\Psi^{(0)}$) and small perturbations ($\rho^{(1)} =: \rho'$ resp. $\Psi^{(1)} =: \psi$).
7. The quantities ρ, λ, μ, g, and Ψ depend solely on r and are independent of the angular coordinates θ and ϕ.

With these assumptions, we can simplify now our equations of motion to the following differential equation system for our displacement modes (for the sake of readability, we have dropped (nml) while still considering only one mode, as in equation 2.28)

I. Linearized Cauchy equation:

$$\mu(u_{k,jj} + (u_{j,j}, k)) + \frac{d\mu}{dr}\left(2\frac{\partial u_k}{\partial r} + \epsilon_{kab}\hat{e}_a^{(r)}(\epsilon_{bcd}u_{d,c})\right) + (\lambda u_{i,i})_{,k}$$
$$+\rho^{(0)}((\psi - g^{(0)}u_r)_{,k} + g^{(0)}\hat{e}_k^{(r)}u_{i,i}) + \omega^2\rho^{(0)}u_k = 0 \tag{2.33}$$

II. Poisson equation for the perturbation of the gravitational potential

$$\psi_{,ii} = 4\pi G(\rho^{(0)}u_j)_{,j} \tag{2.34}$$

A detailed derivation can be found in appendix A. To be able to solve these differential equations numerically, we need to have appropriate boundary conditions:

1. The solution must be well-defined at the origin.
2. Stresses must vanish at the deformed surfaces and remain continuous across any internal deformed surfaces of discontinuity.
3. Displacements must be continuous across internal surfaces of discontinuity, except at a solid-liquid interface where only the radial displacement needs to be continuous.
4. Both the gravitational potential and its radial derivative must be continuous at the deformed surface of the Earth and across any internal deformed surfaces of discontinuity.

To apply these boundary conditions mathematically, we analyze the stresses near a surface of discontinuity at $r = r_c$ in a strained state:

$$\sigma_{jk}(r_c + u_r) = \sigma_{jk}(r_c) + u_r\left(\frac{\partial\sigma_{jk}}{\partial r}\right)_{r=r_c} = \sigma_{jk}^{(e)}(r_c) - p^{(0)}(r_c)\delta_{jk}. \tag{2.35}$$

Here, the additional elastic stresses at r_c and $r_c + u_r$ are equal to first order in u_r. A small element carries its initial stress as it moves. Boundary condition 4 concerns only the gravitational potential, expressed as:

$$\Psi_< = \Psi_> \quad \text{and} \quad \left.\frac{d\Psi_<}{dr}\right|_{r=r_c+u_r} = \left.\frac{d\Psi_>}{dr}\right|_{r=r_c+u_r}, \tag{2.36}$$

where Ψ and its derivative are continuous across the surface of discontinuity. Index $>$ and $<$ represent Ψ being evaluated at opposite points of the discontinuity-surface, which will later on be the surface between the mantle

and the core. Expanding Ψ around its equilibrium value and using Poisson's equation we get

$$\frac{d^2\Psi^{(0)}}{dr^2} + \frac{2}{r}\frac{d\Psi^{(0)}}{dr} = -4\pi G\rho^{(0)}. \tag{2.37}$$

We can furthermore set restrictions on the deviation ψ of the graviational potential:

$$\psi_< = \psi_>, \quad \dot{\psi}_< - 4\pi G\rho_<^{(0)} u_r = \dot{\psi}_> - 4\pi G\rho_>^{(0)} u_r. \tag{2.38}$$

At the Moon's surface, we then have:

$$\psi = \psi^{(e)}, \quad \dot{\psi} - 4\pi G\rho^{(0)} u_r = \dot{\psi}^{(e)}, \tag{2.39}$$

where $\psi^{(e)}$ is the gravitational potential outside the spherical model.

3

Toroidal Oscillations

This chapter summarizes the results of [1, 10, 14]. We will derive a differential equation system suitable for numerical integration that describes purely toroidal oscillations. Starting from the simplified linearized Cauchy equation, we express the toroidal displacement in spherical coordinates and use the boundary conditions from chapter 2 to transform this problem into a system of first-order linear differential equations, which can be numerically solved. Next, we develop the toroidal Green tensor, enabling us to express displacements induced by force distributions in terms of Green's functions. Finally, we use these tools to describe the toroidal displacements induced by gravitational waves.

3.1 Differential Equation System for Numerical Integration

For purely toroidal oscillations u_r and $u_{k,k}$ both vanish. The linearized Cauchy equation then simplifies to

$$\mu u_{j,ii} + \frac{d\mu}{dr}\left(2\frac{\partial u_j}{\partial r} + \epsilon_{jab}\hat{e}_a^{(r)}(\epsilon_{bcd}u_{d,c})\right) + \omega^2\rho^{(0)}u_j = 0. \tag{3.1}$$

As we already mentioned in section 2.4, toroidal oscillations have displacement-modes in the form of $V(r) = W(r) = 0$ in equation 2.28. With this in mind, we first split the displacement into core and mantle contributions, $\sigma = \{c, s\}$, ensuring the solution is well-defined at the origin and throughout the mantle. Using the ansatz from equation 2.28 we get

$$u_j(\mathbf{r}) = \sum_{\sigma,m,l} y_1(r)\sqrt{l(l+1)}C_j^{(\sigma ml)}(\vartheta, \varphi). \tag{3.2}$$

Inserting this Ansatz into 3.1 we get

$$\mu\left(\frac{d^2y_1}{dr^2}+\frac{2}{r}\frac{dy_1}{dr}\right)+\frac{d\mu}{dr}\left(\frac{dy_1}{dr}-\frac{y_1}{r}\right)+\omega^2\rho^{(0)}y_1-\frac{l(l+1)}{r^2}\mu y_1=0. \tag{3.3}$$

By boundary condition 2, stresses must vanish at the core-mantle boundary. The radial stress is written as

$$\hat{e}_i^{(r)}\sigma_{ij}=\sum_{\sigma,m,l}y_2(r)\sqrt{l(l+1)}C_j^{(\sigma ml)}(\vartheta,\varphi), \tag{3.4}$$

where we defined a function y_2 that is zero at the boundary. Inserting the displacement equation into the stress tensor we find

$$y_2=\mu\left(\frac{dy_1}{dr}-\frac{y_1}{r}\right). \tag{3.5}$$

Boundary condition 2 requires now, that the stress must vanish at the core radius r and the radius of the Moon R:

$$y_2(r)=0 \quad \text{at} \quad r=r_c \quad \text{and} \quad r=R. \tag{3.6}$$

To avoid numerical issues caused by the derivative $d\mu/dr$, we can use the following equivalent system of two simultaneous linear differential equations of the first order in y_1 and y_2

$$\begin{aligned}\frac{dy_1}{dr}&=\frac{y_1}{r}+\frac{y_2}{\mu},\\ \frac{dy_2}{dr}&=\left(\frac{l^2+l-2}{r^2}\mu-\omega^2\rho^{(0)}\right)y_1-\frac{3}{r}y_2.\end{aligned} \tag{3.7}$$

These differential equations can be now solved numerically under the given boundary conditions (see 6.7 in [14]).

3.2 Toroidal Green Tensor

The displacement u_i induced by the force distribution f_i and the surface stresses σ_{ij} in a radially heterogeneous, anelastic self-gravitating Moon model can be written with a Green's function of the Cauchy equation of motion. If we find this Green's function, the displacement mode is given by the following equation:

$$u_i^{(nml)}(r) = \int_V \mathcal{G}_{ji}^{(nml)}(\mathbf{r}|\mathbf{r}', t) f_j^{(n)}(\mathbf{r}')\, d^3r' + \int_S \mathcal{G}_{ji}^{(nml)}(\mathbf{r}|\mathbf{r}', t)\hat{e}_k \sigma_{kj}^{(0)}\, dS' \quad (3.8)$$

Let us now derive $\mathcal{G}_{ji}^{(nml)}$. We denote the eigenvectors of the Moon's toroidal oscillations corresponding to the eigenfrequency ω_n by $M^{(nml)}$. These eigenvectors form a complete orthogonal basis

$$\frac{1}{\Lambda_T^{(nml)}} \int_V M^{(nml)}(r) \cdot M^{*(\tilde{n}ml)}(r) \rho^{(0)}(r)\, dV = \delta_{n\tilde{n}}, \quad (3.9)$$

where the integral spans the entire Moon's volume and $\Lambda_T^{(nml)}$ is the normalization factor. The toroidal displacement can be expanded in terms of these eigenvectors. One mode is then given by

$$u_i^{(nml)}(r) = a^{(nml)} M_i^{(nml)}(r), \quad (3.10)$$

with

$$a^{(nml)} = \frac{1}{\Lambda_T^{(nml)}} \int_V u_i^{(nml)} \cdot M_i^{*(nml)} \rho_0\, dV. \quad (3.11)$$

Substituting Eq. 3.10 into the general Fourier-transformed equation of motion 2.19 we obtain

$$a^{(nml)} \sigma_{ij,j}^{(0)} + \rho^{(0)} F_i + \rho^{(0)} \omega^2 a_i^{(n)} M_i^{(nml)} = 0. \quad (3.12)$$

By taking the scalar product of equation 3.12 with $M^{*(nml)}$, integrating over V and using the orthogonality of $M^{(nml)}$ we arrive at

$$\int_V a^{(nml)} \sigma_{ij,j}^{(0)} M^{*(nml)}\, dV + \int_V M^{*(nml)} \rho^{(0)} F_i\, dV + \omega^2 \Lambda_T^{(nml)} a^{(nml)} = 0. \quad (3.13)$$

The eigenvector $M_i^{(\tilde{n}ml)}$ should also satisfy the equation of motion with $F_i = 0$, which implies

$$\sigma_{ij,j}^{(0)} + \rho^{(0)} \omega_k M_i^{(\tilde{n}ml)} = 0. \quad (3.14)$$

By taking the scalar product of this equation with $M_i^{*(nml)}$, integrating over V, and using the orthogonality equation, we obtain

$$\int_V \sigma^{(0)}_{ij,j} M^{*(nml)}\, dV + \omega^2_{\tilde{n}} \delta_{n\tilde{n}} \Lambda_T^{(nml)} = 0. \tag{3.15}$$

Substituting this result into equation 3.13, we derive

$$a^{(nml)} = \frac{1}{(\omega_n^2 - \omega^2)\Lambda_T^{(nml)}} \int_V M_i^{*(nml)} F_i \rho^{(0)}\, dV. \tag{3.16}$$

Thus, our displacement can be reformulated as

$$u_i^{(nml)}(r) = \frac{1}{(\omega_n^2 - \omega^2)\Lambda_T^{(nml)}} \int_V M_i^{(nml)} M_j^{*(nml)} F_j \rho^{(0)}\, dV. \tag{3.17}$$

Time-dependence is given by equation 2.21:

$$u_i^{(nml)}(r,t) = \frac{1}{(\omega_n^2 - \omega^2)\Lambda_T^{(nml)}} \int_V M_i^{(nml)} M_j^{*(nml)} F_j \rho^{(0)}\, dV \bar{g}^{(n)}(t). \tag{3.18}$$

Comparing this result with equation 3.8 we finally found the toroidal Green's function

$$\mathcal{G}_{ij}^{(nml)} = \frac{1}{\Lambda_T^{(nml)}} M_i^{*(nml)}(r) M_j^{(nml)}(r^{(0)}) \bar{g}^{(n)}(t). \tag{3.19}$$

Since we only look at toroidal displacements, $M_i^{(nml)}$ takes the form described in equation 3.2:

$$M_i^{(nml)} = (y_1^{(n)}(r))^T \sqrt{l(l+1)} C_i^{ml}(\theta, \phi) \tag{3.20}$$

From the orthogonality we can then derive (see [14] for detailed derivation)

$$\Lambda_T^{(nml)} = 4\pi \frac{l(l+1)}{2l+1} \int_0^R \left[(y_1^{(n)}(r))^T \right]^2 \rho^{(0)} r^2\, dr \tag{3.21}$$

In 3.19 we dropped the term $(\omega_n^2 - \omega^2)$. Since we physically do not want to model a perfect harmonic oscillator, we want to be able to introduce a damping term in $\bar{g}^{(n)}(t)$. The term is therefore included in $\bar{g}^{(n)}(t)$ and the time-dependence is given by

$$\bar{g}^{(n)}(t) = \int \frac{d\omega}{2\pi} \frac{g(\omega) e^{i\omega t}}{\omega_n^2 - \omega^2 + i\omega_n \omega / Q_n} \tag{3.22}$$

where $g(\omega)$ is the Fourier transformation of $g(t)$ and Q_n is the damping factor of the mode n. As a simple model for $g(t)$, we can for example consider an infinite monochromatic wave

$$g(t) = e^{i\omega_0 t}. \tag{3.23}$$

The Fourier transform is then given by

$$g(\omega) = \frac{1}{2\pi}\int e^{i(\omega_0-\omega)t}\,dt = \delta(\omega_0-\omega). \tag{3.24}$$

Taking the model of the damped harmonic oscillator we arrive at

$$\begin{aligned}\bar{g}^{(n)}(t) &= \frac{1}{2\pi}\int d\omega\delta(\omega_0-\omega)\frac{e^{i\omega t}}{\omega_n^2-\omega^2+i\omega_n\omega/Q_n}\\ &= \frac{1}{2\pi}\frac{e^{i\omega_0 t}}{\omega_n^2-\omega_0^2+i\omega_0\omega_n/Q_n}.\end{aligned} \tag{3.25}$$

3.3 Induced toroidal motion from gravitational waves

Combining equation 3.8 with our Green's function and the force coming from the graviational wave, we can rewrite this expression to

$$\begin{aligned}u_i^{(nml)}(r,t) = &-\int_V \frac{d\mu}{dr}(r)\mathcal{G}_{ij}^{(nml)}(r\mid r',t)\hat{e}_k^{(r)}h^{kj}d^3r' + \mu(R)\int_S \mathcal{G}_{ij}^{(nml)}(r\mid r',t)\hat{e}_k^{(r)}h^{kj}dS'\\ &-\frac{1}{2}\int_V \frac{d\lambda}{dr}(r)\mathcal{G}_{ij}^{(nml)}(r\mid r',t)\hat{e}_j^{(r)}hd^3r' + \frac{1}{2}\lambda(R)\int_S \mathcal{G}_{ij}^{(nml)}(r\mid r',t)\hat{e}_j^{(r)}hdS'.\end{aligned} \tag{3.26}$$

We can simplify the expression for $u_i^{(nml)}(r,t)$ as follows:

$$u_i^{(nml)}(r,t) = h_0(\Lambda_T^{(nml)})^{-1}\bar{g}(t)y_1^{(n)}(r)\sqrt{l(l+1)}C_i^{*(lm)}(\theta,\phi)F_T \tag{3.27}$$

where

$$\begin{aligned}F_T = &R^2y_1^{(n)}(R)\left[\mu(R)\mathcal{E}^{ij}+\frac{1}{2}\lambda(R)\delta_{ij}\right]\widetilde{C}_{ij}^{(lm)}(kR)\\ &-\int_0^R dr\,r^2y_1^{(n)}(r)\left[\frac{d\mu(r)}{dr}\mathcal{E}^{ij}+\frac{1}{2}\frac{d\lambda(r)}{dr}\delta_{ij}\right]\widetilde{C}_{ij}^{(lm)}(kr),\end{aligned} \tag{3.28}$$

and

$$\widetilde{C}_{ij}^{(lm)}(kR) := \int_0^{2\pi}d\phi\int_0^{\pi}d\theta\sin\theta\hat{e}_i^{(r)}C_j^{(lm)}e^{-i\mathbf{kr}}. \tag{3.29}$$

The polarization tensor $\mathcal{E}^{ij}$ is taken from the introduction. From 3.28 it is clear that F_T can be written as a sum of a μ and a λ dependent part: $F_T = F_T^{(\mu)} + F_T^{(\lambda)}$. Performing a partial-wave expansion of $\tilde{C}_{ij}^{(lm)}(kR)$ (where we only keep the partial waves with the lowest angular momentum) and using the low frequency limit $kr \to 0$ proposed by [10], we can expand F_T in terms of Wigner symbols.

This calculation is quite lengthy and will not be covered in this review. Readers are encouraged to read the derivation by Ben-Menahem [14] for more details. We get

$$
\begin{aligned}
F_T^{(\mu)} =& 8\pi i \sum_{l_1=0}^{\infty} \sum_{m_l=-l_1}^{l_1} i^{-l_1} \frac{(2l_1+1)}{(2l_1+1)!!} \sqrt{(l+m)(l-m+1)} \\
&\times \sqrt{\frac{(l_1-m_1)!}{(l_1+m_1)!}} P_{l_1}^{m_1}(1) \begin{pmatrix} l_1 & l & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & l & 1 \\ m_1 & m-1 & -1 \end{pmatrix} \\
&\times \left[\mu(R) R^2 (y_1^{(n)}(R))^T (kR)^{l_1} - \int_0^R \frac{d\mu}{dr} (y_1^{(n)}(r))^T (kr)^{l_1} r^2 dr \right]
\end{aligned}
\tag{3.30}
$$

where

$$
\begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \tag{3.31}
$$

are the Wigner symbols. A brief introduction to Wigner symbols can be found in appendix B.

It can be shown [14], that in the low-frequency limit the $F_T^{(\lambda)}$ part vanishes, and we get $F_T = F_T^{(\mu)}$. Therefore scalar polarized waves do not significantly contribute to toroidal displacements.

4

Spheroidal Oscillations

In this chapter, we closely follow again [1, 10, 14]. We will, analogous to the last chapter, find expressions for the radial functions $y_p^{(n)}$, now for spheroidal displacements and gravitational perturbations. Next, we will address the induced spheroidal motion caused by gravitational waves, deriving the Green's Tensor for spheroidal displacements. Finally, we will explore the quadrupole moment of the Moon, leading to expressions that simplify the spheroidal contributions to the Moon's deformation.

4.1 Differential Equation System for Numerical Integration

We will follow the same procedure as discussed in chapter 3. Spheroidal oscillations are described by displacements involving only the vector surface harmonics $P_i^{(ml)}$ and $B_i^{(ml)}$. Hence, a spheroidal displacement can be expressed as:

$$u_j(\mathbf{r}) = \sum_{\sigma,m,l} \left(y_1^{(n)}(r) P_j^{(\sigma ml)}(\vartheta, \phi) + y_3^{(n)}(r) \sqrt{l(l+1)} B_j^{(\sigma ml)}(\vartheta, \phi) \right) \tag{4.1}$$

where again $\sigma \in \{c, s\}$ to split the displacement into core and mantle contributions. We now furthermore decompose the gravitational perturbation as

$$\psi(\mathbf{r}) = \sum_{\sigma,m,l} y_5^{(n)}(r) Y^{(\sigma ml)}(\vartheta, \phi). \tag{4.2}$$

Inserting equations 4.1 and 4.2 into our differential equation system in 2.5 we can set

$$X = \frac{dy_1}{dr} + \frac{2}{r}y_1 - \frac{l(l+1)}{r}y_3, \quad \text{and} \quad Z = \frac{1}{r}(y_1 - y_3) - \frac{dy_3}{dr} \tag{4.3}$$

and after equating to zero the coefficients of $P_i^{(ml)}$ and $B_i^{(ml)}$ in the equations thus obtained, we find

$$\mu\left(2\frac{dX}{dr} - \frac{l(l+1)}{r}Z\right) + \frac{d(\lambda X)}{dr} + 2\dot{\mu}\frac{dy_1}{dr} + \rho^{(0)}\left[\frac{dy_5}{dr} - 4\pi G\rho^{(0)}y_1 + g^{(0)}\left(X - \frac{dy_1}{dr} + \frac{2}{r}y_1 + \omega^2 y_1\right)\right] = 0, \tag{4.4}$$

$$(\lambda + 2\mu)\frac{X}{r} - \frac{d}{dr}(\mu Z) - \mu\frac{Z}{r} + 2\dot{\mu}\left(\frac{dy_3}{dr} + Z\right) + \rho^{(0)}\left(\frac{1}{r}\left(y_5 - g^{(0)}y_1\right) + \omega^2 y_3\right) = 0. \tag{4.5}$$

Again we define y_2 and y_4 such that

$$\hat{e}_i^{(r)}\sigma_{ij} = \sum_{\sigma,m,l}\left(y_2(r)P_j^{(\sigma ml)}(\vartheta,\phi) + y_4(r)\sqrt{l(l+1)}B_j^{(\sigma ml)}(\vartheta,\phi)\right) \tag{4.6}$$

we find y_2 and y_4 by inserting 4.1 into equation 4.6:

$$\begin{aligned} y_2 &= \lambda X + 2\mu\frac{dy_1}{dr} = (\lambda + 2\mu)\frac{dy_1}{dr} + \frac{2\lambda}{r}y_1 - \lambda\frac{l(l+1)}{r}y_3, \\ y_4 &= \mu\left(Z + 2\frac{dy_3}{dr}\right) = \mu\left(\frac{1}{r}(y_1 - y_3) + \frac{dy_3}{dr}\right). \end{aligned} \tag{4.7}$$

We now insert 4.2 into the second equation in 2.5 to obtain a differential equation for y_5:

$$\frac{d^2y_5}{dr^2} + \frac{2}{r}\frac{dy_5}{dr} - \frac{l(l+1)}{r^2}y_5 = 4\pi G\left(\rho^{(0)}X + \dot{\rho}^{(0)}y_1\right). \tag{4.8}$$

If we take the value of ψ outside the boundary of the spherical model then by $\Delta\psi_{(e)} = 0$ we can write the boundary condition as

$$\frac{dy_5}{dr} - 4\pi G\rho^{(0)}y_1 = -\frac{l+1}{r}y_5 \quad \text{at} \quad r = R. \tag{4.9}$$

Defining a new function y_6 as

$$y_6 = \frac{dy_5}{dr} - 4\pi G\rho^{(0)}y_1 \tag{4.10}$$

the boundary conditions at $r = R$ become

$$y_2 = 0, \quad y_4 = 0, \quad y_6 + \frac{l+1}{r} y_5 = 0. \tag{4.11}$$

These are all the ingredients we need, to obtain a system of first-order differential equations for $y_1, y_2, \ldots, y_6$ valid in the mantle:

$$\frac{dy_1}{dr} = -\frac{2\lambda}{(\lambda+2\mu)r} y_1 + \frac{1}{\lambda+2\mu} y_2 + \frac{l(l+1)\lambda}{(\lambda+2\mu)r} y_3 \tag{4.12}$$

$$\begin{aligned} \frac{dy_2}{dr} = & \left[-\omega^2 \rho^{(0)} - 4\frac{g^{(0)}\rho_0}{r} + \frac{4\mu(3\lambda+2\mu)}{(\lambda+2\mu)r^2} \right] y_1 - \frac{4\mu}{(\lambda+2\mu)r} y_2 \\ & + \frac{l(l+1)}{r} \left[g^{(0)}\rho^{(0)} - \frac{2\mu(3\lambda+2\mu)}{(\lambda+2\mu)r} \right] y_3 + \frac{l(l+1)}{r} y_4 - \rho^{(0)} y_6 \end{aligned} \tag{4.13}$$

$$\frac{dy_3}{dr} = -\frac{1}{r} y_1 + \frac{1}{r} y_3 + \frac{1}{\mu} y_4 \tag{4.14}$$

$$\begin{aligned} \frac{dy_4}{dr} = & \left[\frac{g^{(0)}\rho^{(0)}}{r} - \frac{2\mu(3\lambda+2\mu)}{(\lambda+2\mu)r^2} \right] y_1 - \frac{\lambda}{(\lambda+2\mu)r} y_2 \\ & + \left(-\omega^2 \rho^{(0)} + \left(\left(2l^2 + 2l - 1 \right) \lambda + 2 \left(l^2 + l - 1 \right) \mu \right) \frac{2\mu}{(\lambda+2\mu)r^2} \right) y_3 \\ & - \frac{3}{r} y_4 - \frac{\rho^{(0)}}{r} y_5 \end{aligned} \tag{4.15}$$

$$\frac{dy_5}{dr} = 4\pi G \rho^{(0)} y_1 + y_6 \tag{4.16}$$

$$\frac{dy_6}{dr} = -4\pi \frac{l(l+1)}{r} G \rho^{(0)} y_3 + \frac{l(l+1)}{r^2} y_5 - \frac{2}{r} y_6 \tag{4.17}$$

This system is, as already mentioned, to be used in the mantle. In the core, on the other hand, we have

$$\mu = 0, \quad y_2 = \lambda X, \quad y_4 = 0 \tag{4.18}$$

resulting in:

$$\frac{dy_1}{dr} = -\frac{2}{r}y_1 + \frac{1}{\lambda}y_2 + \frac{l(l+1)}{r}y_3 \tag{4.19}$$

$$\frac{dy_2}{dr} = -\left(\omega^2\rho^{(0)} + \frac{4g^{(0)}\rho^{(0)}}{r}\right)y_1 + \frac{l(l+1)}{r}g^{(0)}\rho^{(0)}y_3 - \rho^{(0)}y_6 \tag{4.20}$$

$$y_3 = \frac{1}{\omega^2 r}\left(g^{(0)}y_1 - \frac{1}{\rho^{(0)}}y_2 - y_5\right) \tag{4.21}$$

$$y_4 = 0 \tag{4.22}$$

$$\frac{dy_5}{dr} = 4\pi G\rho^{(0)}y_1 + y_6 \tag{4.23}$$

$$\frac{dy_6}{dr} = -4\pi\frac{l(l+1)}{r}G\rho^{(0)}y_3 + \frac{l(l+1)}{r^2}y_5 - \frac{2}{r}y_6 \tag{4.24}$$

In both, mantle and core, the differential equations can be now solved numerically (see 6.7 in [14]).

4.2 Induced spheroidal motion from gravitational waves

Analouge to the steps done in section 3.3 to find the Greens Tensor for the displacement of induced toroidal motions, we can also find the Green's Tensor for induced spheroidal motions. The only difference lies in the function $M_i^{(nml)}$, since now the displacement is made up of the vector surface harmonics $P_i^{(nml)}$ and $B_i^{(nml)}$ vectors which means that the Green's Tensor is still given by

$$\mathcal{G}_{ij}^{(nml)} = \frac{1}{\Lambda_S^{(nml)}}Q_i^{*(nml)}(r)Q_j^{(nml)}(r^{(0)})\bar{g}^{(n)}(t) \tag{4.25}$$

but the new eigenvectors take the form

$$Q_i^{(nml)}(r) = y_1^{(n)}(r)P_i^{(ml)}(\theta,\phi) + y_3^{(n)}(r)\sqrt{l(l+1)}B_i^{(ml)}(\theta,\phi). \tag{4.26}$$

The normalization is now given by

$$\Lambda_S^{(nml)} = \frac{4\pi}{2l+1}\int_0^R ((y_1^{(n)})^2 + l(l+1)(y_3^{(n)})^2)\rho^{(0)}(r)r^2dr. \tag{4.27}$$

Again using the form in equation 3.8 and inserting the Green's Tensor we get an expression for the induced displacement from the spheroidal oscillations:

$$u_j^{(nml)}(\boldsymbol{r},t) = h_0\bar{g}(t)\left(\Lambda_S^{nml}\right)^{-1}\left[P_j^{*(nml)}(\boldsymbol{r})F_{S_1} + \sqrt{l(l+1)}B_j^{*(nml)}(\boldsymbol{r})F_{S_2}\right], \tag{4.28}$$

where

$$\begin{aligned} F_{S_1} &= R^2\mu(R)y_1^{(n)}(R)\mathcal{E}^{ij}\widetilde{P}_{ij}^{(lm)}(kR) - \int_0^R \frac{d\mu}{dr}y_1^{(n)}(r)r^2dr\mathcal{E}^{ij}\widetilde{P}_{ij}^{(lm)}(kr) \\ F_{S_2} &= R^2\mu(R)y_1^{(n)}(R)\mathcal{E}^{ij}\widetilde{B}_{ij}^{(lm)}(kR) - \int_0^R \frac{d\mu}{dr}y_1^{(n)}(r)r^2dr\mathcal{E}^{ij}\widetilde{B}_{ij}^{(lm)}(kr) \end{aligned} \tag{4.29}$$

and again

$$\widetilde{P}_{ij}^{(lm)}(kR) := \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta \hat{e}_i^{(r)} P_j^{(lm)} e^{-i\mathbf{kr}} \tag{4.30}$$

(analogously for $\widetilde{B}_i^{(lm)}$). Terms including the Lamé parameter λ do not contribute to F_S. For further discussions on this, see [1]. In order now to simplify the expressions for F_{S_1} and F_{S_2}, we will study the quadrupole moment $\mathcal{D}_{ij}$ of the Moon

$$\mathcal{D}_{ij} = 3\int_V r_i r_j \rho dV. \tag{4.31}$$

Since we are only interested in the change of the quadrupole moment after a displacement $r_i \to r_i + u_i$, we take the first order perturbation of $\mathcal{D}$ which gives us

$$\delta\mathcal{D}_{ij} = 3\int_V (r_i u_j + u_i r_j)\rho dV. \tag{4.32}$$

Inserting only a single spheroidal mode

$$u_i^{nml} = y_1^{(n)}(r)P_i^{(ml)}(\theta,\phi) + y_3^{(n)}(r)\sqrt{l(l+1)}B_i^{(ml)}(\theta,\phi) \tag{4.33}$$

we get

$$\delta\mathcal{D}_{ij} = \delta\mathcal{D}_{ij}^{(B)} + \delta\mathcal{D}_{ij}^{(P)} \tag{4.34}$$

with

$$\begin{aligned} \delta\mathcal{D}_{ij}^{(P)} &= 3\int_0^R y_1^{(n)}(r)\rho^{(0)}(r)r^2dr\left(\widetilde{P}_{ij}^{(ml)}(r) + \widetilde{P}_{ji}^{(ml)}(r)\right), \\ \delta\mathcal{D}_{ij}^{(B)} &= 3\sqrt{l(l+1)}\int_0^R y_1^{(n)}(r)\rho^{(0)}(r)r^2dr\left(\widetilde{B}_{ij}^{(ml)}(r) + \widetilde{B}_{ji}^{(ml)}(r)\right). \end{aligned} \tag{4.35}$$

We can express now F_{S_1} and F_{S_2} in terms of $\delta\mathcal{D}$:

$$\begin{aligned} F_{S_1} &= \frac{R^2\mu(R)y_1^{(n)}(R) - \int_0^R \dot{\mu}y_1^{(n)}(r)r^2dr}{3\int_0^R \rho_0 y_1^{(n)}(r)r^2dr}\mathcal{E}^{ij}\delta\mathcal{D}_{ij}^{(P)}, \\ F_{S_2} &= \frac{R^2\mu(R)y_1^{(n)}(R) - \int_0^R \dot{\mu}y_1^{(n)}(r)r^2dr}{3\int_0^R \rho_0 y_1^{(n)}(r)r^2dr}\mathcal{E}^{ij}\delta\mathcal{D}_{ij}^{(B)}, \end{aligned} \tag{4.36}$$

and the notation $\dot{\mu} := \frac{d\mu}{dr}$ is used. Following now once again the low frequency approximation $kr \to 0$, one gets

$$\delta\mathcal{D}_{ij} = \frac{24\pi}{5\sqrt{6}}\delta_{l,2}\mathbf{\Delta}_{ij}\int_0^R (y_1^{(n)}(r) + 3y_3^{(n)}(r))\rho^{(0)}(r)r^2 dr, \tag{4.37}$$

where the tensor $\mathbf{\Delta}_{ij}$ is explicitely given by

$$\mathbf{\Delta}_{ij} = \begin{pmatrix} \delta_{m,2} + \delta_{m,-2} - \sqrt{\frac{2}{3}}\delta_{m,0} & \mathrm{i}\delta_{m,2} - \mathrm{i}\delta_{m,-2} & \delta_{m,1} + \delta_{m,-1} \\ \mathrm{i}\delta_{m,2} - \mathrm{i}\delta_{m,-2} & -\delta_{m,2} - \delta_{m,-2} - \sqrt{\frac{2}{3}}\delta_{m,0} & -\mathrm{i}\delta_{m,-1} + \mathrm{i}\delta_{m,1} \\ \delta_{m,1} + \delta_{m,-1} & -\mathrm{i}\delta_{m,-1} + \mathrm{i}\delta_{m,1} & 2\sqrt{\frac{2}{3}}\delta_{m,0} \end{pmatrix}. \tag{4.38}$$

The expression in equation 4.37 contains $\delta_{l,2}$ and therefore only spheroidal oscillations with $l = 2$ will contribute in this approximation.

5

Moon Response

In this chapter, we will summarize the results from papers [1] and [15]. We will discuss the calculation of eigenfrequencies for different modes and compare the results across three Moon models. Following this, we will examine the contributions of toroidal and spheroidal modes to the Moon's oscillations. Finally, we will analyze the frequency-dependent response of the Moon to gravitational waves, highlighting key findings and implications for detectability.

5.1 Moon model

As we have seen in the previous chapters, our Moon model is completely described by the density $\rho(r)$, the Lamé parameters $\lambda(r)$ and $\mu(r)$, and gravitational acceleration $g(r)$. All these values have to be determined experimentally. For a detailed explanation of the model construction, please refer to reference [15]. The parameters of the models are shown in Figure 5.1. All models predict a similar behavior of all parameters throughout most of the mantle, at least to a certain approximation. However, in the core, the parameters for all models are very different. Especially, the Lamé parameter $\lambda(r)$ splits at the core into three different values.

5.2 Eigenfrequencies

By following the procedures discussed in the last few chapters, one can determine the eigenfrequencies of the Moon models. Measuring the eigenfrequencies for different modes reveals that toroidal eigenfrequencies show minimal variation between models, as these oscillations occur primarily in

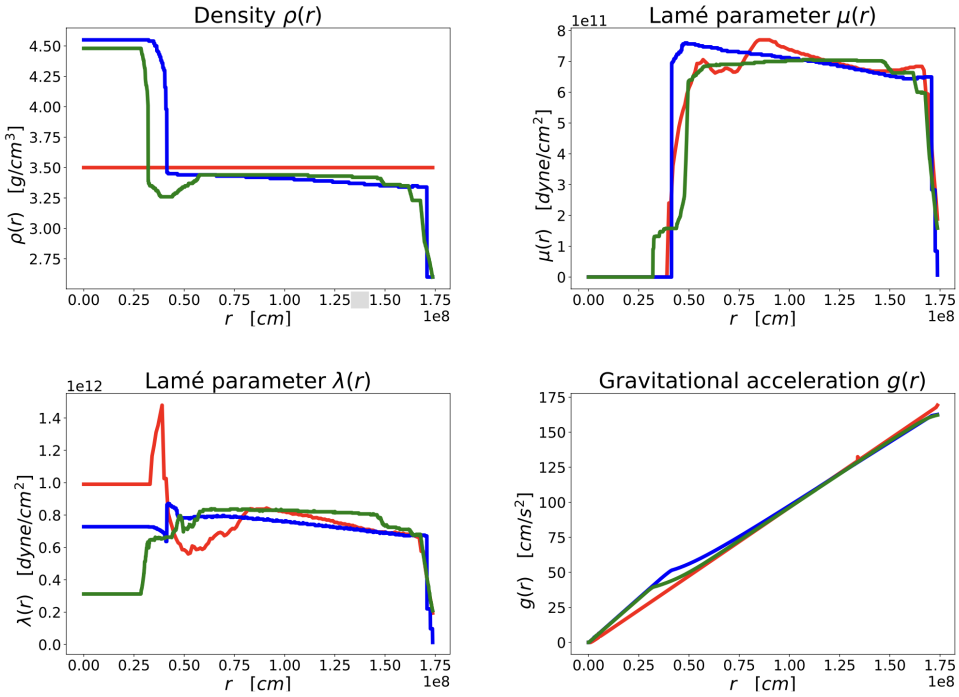

Figure 5.1: *The density $\rho(r)$, the Lamé parameters $\lambda(r)$ and $\mu(r)$, and gravitational acceleration $g(r)$ of three Moon models taken from [15]. Moon model 1 in red, model 2 in blue and model 3 in green*

n	Model 1	Model 2	Model 3
0	0.993	1.014	1.011
1	2.901	2.950	2.918
2	4.266	4.388	4.079
3	5.635	5.869	4.929
0	1.098	1.050	1.047
1	1.848	1.890	1.892
2	2.932	2.881	2.688
3	3.976	3.993	3.412

Table 5.1: *First four eigenfrequencies $\nu = \omega/2\pi$ the three Moon models in mHz (toroidal on top and spheroidal on bottom)*

the mantle and crust where the models largely agree. However, spheroidal oscillations show more significant differences, since the core significantly influences these frequencies. The first four eigenfrequencies calculated for the three different Moon models by Kachelriess [1] are shown in Table 5.1. Although the variation of the eigenfrequencies between the three models is already very high for $n = 4$, they agree very well in the fundamental eigenfrequencies.

5.3 Modes and Displacements

Before we begin to show the results of the elastic displacements of the Moon, we should take a look, which toroidal modes ${}_mT_l$ and spheroidal modes ${}_mS_l$ contributes the most to the oscillations.

5.3.1 Toroidal Modes

The biggest contribution in 3.30 should come from $l_1 = 0$. This term, however, turns out to be zero. The $l_1 = 1$ contribution however is non-zero for $l = 2$, since the Wigner symbols must require that $l + l_1$ is odd to get a nonzero contribution. Inserting this into equation 3.30 we see that the Wigner symbol

$$\begin{pmatrix} 1 & 2 & 1 \\ m_1 & m-1 & -1 \end{pmatrix} \tag{5.1}$$

is non-zero, if $m_1 + m - 1 - 1 = 0$. Since $-2 \leq m \leq 2$ and $-1 \leq m_1 \leq 1$, we see that $m = 2$ and $m = 1$ are the only possible candidates. But since the Legendre polynomial P_2^1 vanishes, we get $m = 2$ as the only contribution that survives for $l = 2$ and therefore the only mode interesting to us is ${}_2T_2$.

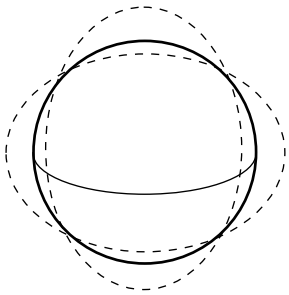

Figure 5.2: *Normal mode oscillation ${}_0S_2$. This type of mode is only triggered, when the GW has scalar polarization*

5.3.2 Spheroidal Modes

We recall from the last chapter, that $l = 2$ in the case of spheroidal oscillations. In this case, $l_1 = 0$ already gives a non-zero result. It is important to mention, that $l_1 = 0$ has already been used in 4.37 to get a simpler form. From the quadrupole tensor $\mathbf{\Delta}_{ij}$ it is clear, that the only contributions are $m = 0$ (if we have a scalar polarization A_S) or $m = 2$ (for the cross and plus polarizations) taking the contraction with $\mathcal{E}_{ij}$. Therefore the only modes interesting to us are ${}_0S_2$ and, especially, ${}_2S_2$. An example of the normal mode oscillation ${}_0S_2$ is illustrated in figure 5.2.

5.3.3 Displacements at resonance

Displacements of spherical bodies can be written as a product of a radial and an angular dependent part. We will neglect now the angular-dependent spherical vector functions and just focus on the overall magnitude of the mode

$$\begin{aligned} \xi_T(t) &= h_0 \left(\Lambda_T^{(022)}\right)^{-1} F_T \bar{g}(t), \\ \xi_S(t) &= h_0 \left(\Lambda_S^{(022)}\right)^{-1} (F_{S_1} + F_{S_2})\, \bar{g}(t). \end{aligned} \tag{5.2}$$

We will only consider displacements at resonance, which mathematically just means $\omega_n = \omega_0$. The value $Q_1 = 3300$ as a damping factor is used, following [15]. Setting $h_0 = 1$ and assuming $A_+ = A_\times = 1$, one gets the values presented in table 5.2. Comparing this result to other papers discussing the Earth response to GW [16–18], Moon displacements show a higher response by a factor of two orders of magnitude. When comparing the toroidal values of model 1 and model 3, there is good agreement. Model 2 nevertheless has a significantly higher value compared to the two other models. According to the model Lamé parameter plot in Figure 5.1, model 2 differs notably near

Mode	Model 1	Model 2	Model 3
${}_2T_2$	7.127×10^6cm	1.530×10^7cm	7.520×10^6cm
${}_2S_2$	3.261×10^{11}cm	5.642×10^{11}cm	4.814×10^{11}cm
${}_0S_2$	2.663×10^{11}cm	4.606×10^{11}cm	3.931×10^{11}cm

Table 5.2: *Displacements ξ_T/h_0 and ξ_S/h_0 for different moon models and modes with leading contributions.*

the surface, particularly in the second Lamé parameter $\mu(r \approx R)$, which is two orders of magnitude lower than that of models 1 and 3. This discrepancy in $\mu(r)$ may account for the difference. In the spheroidal case, a similar trend is observed as with the toroidal mode: model 2 has a greater response than the other models, with model 3 slightly exceeding model 1. However, the differences are much lower in the spheroidal case. This is likely due to the fact, that the spheroidal mode depends on the Lamé parameter values of both the core and the mantle, unlike the toroidal mode which depends primarily on the mantle. At last, focusing on the mode excited by the scalar polarized gravitational wave ${}_0S_2$, we notice a similar pattern as the ${}_2S_2$ mode. Model 2 again shows the highest response, followed by model 3, with model 1 displaying the weakest response. Additionally, we observe a significant smaller contribution for this mode in the same models as the ${}_2S_2$ mode. This can also be shown mathematically. If we contract 4.38 only with the scalar polarization, we get a contribution of

$$A_s\left(\delta_{m,2} + \delta_{m,-2} - \sqrt{\frac{2}{3}}\delta_{m,0}\right) + A_s\left(-\delta_{m,2} - \delta_{m,-2} - \sqrt{\frac{2}{3}}\delta_{m,0}\right) = -2A_s\sqrt{\frac{2}{3}}. \tag{5.3}$$

However, if we take the cross and plus polarization, we get a contribution of

$$2A_+\left(\delta_{m,2} + \delta_{m,-2}\right), \quad \text{and} \quad 2\mathrm{i}A_\times\left(\delta_{m,2} - \delta_{m,-2}\right) \tag{5.4}$$

in Einstein gravity. This is precisely the factor $\sqrt{\frac{2}{3}}$ difference we see in the data given by table 5.2.

5.4 Frequency-Dependent Response

Until now, our data has been only for the first eigenfrequency of the Moon models, focusing solely on the response corresponding to this frequency. However, since gravitational waves (GW) typically do not align precisely with

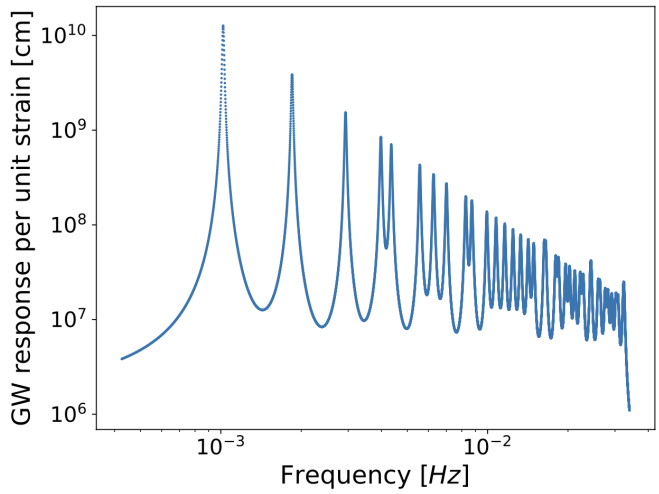

Figure 5.3: *Gravitational wave response in Moon model 1 including the eigenfrequencies* $n = 0, 1, \ldots, 36$ *with* $Q_n = 3300$.

the Moon's eigenfrequency and may have a broad spectrum of frequencies, it becomes interesting to investigate how the response varies across different frequencies ω_n. We still assume a gravitational wave travelling in z-direction with frequency ω_0. Since the contribution of spheroidal oscillations are much bigger, we will neglect toroidal oscillations for now, so

$$\xi_S(t) = \left(\Lambda_S^{(n22)}\right)^{-1} (F_{S_1} + F_{S_2})\, \bar{g}(t). \tag{5.5}$$

As one can see, we restrict ourselves still to $l = 2$ and $m = 2$ modes, since they have the highest contributions. In total we have now a sum over all frequencies and want to know, what the function

$$\xi_{\text{tot.}}(t) = \sum_{n=0}^{\infty} \xi^{n22}(t) \tag{5.6}$$

looks like. We therefore make the assumption that the response from all the eigenfrequencies adds constructively. Figure 5.3 shows the GW response per unit strain at different frequencies. In Kachelriess paper [1], the damping factor $Q_n = 3300$ is chosen to be constant for all n. Although there is a paper [19] about the dependence of Q on frequencies in $3-8$Hz range, the range mHz-Hz has not been covered yet in any other paper. Since in [19], Q_n has almost been found to be constant, at least for S waves, this is a good approximation for Q_n. We observe that as the frequency increases, the response generally decreases. Additionally, the gap between resonance frequencies narrows

at higher frequencies, indicating the Moon can pick up gravitational waves from a large span of different frequencies, especially at higher frequencies $\omega \gg \omega_1$. For frequencies below the first eigenfrequency, the response is strongly suppressed, making it unlikely to detect gravitational waves below the mHz range. Figure 5.3 shows that above approximately 0.03Hz, the response is suppressed due to missing resonances, causing a notable decrease at the highest frequencies. Using the reach in measuring accelerations predicted for the Lunar gravitational wave antenna (LGWA) experiment from Harms [20], Kachelriess finally determines that this experiment has a nominal sensitivity to gravitational waves with an amplitude of approximately

$$h_0 \approx 10^{-18} \text{ in the mHz range.}$$

6 Conclusion

In this report, we studied the response of the Moon to gravitational waves in both general relativity and Brans-Dicke gravity. Our calculations in this report indicated a maximum response of approximately $h_0 \times 10^{11}$ cm for the amplitude h_0 of gravitational waves. Spheroidal displacements are significantly higher than toroidal displacements due to a dominant first term in the series expansion of the displacement. The response of a spherical body to gravitational waves depends on the frequency of the incoming wave, with lower frequencies showing resonance behavior and higher frequencies showing broadband behavior. The Moon's normal modes are most sensitive in the gravitational wave frequency band of $\nu = \omega/2\pi \approx 1.0 \pm 0.02$ mHz.

Current knowledge of the Moon's interior is limited, and placing Lunar gravitational wave antennas could improve model accuracy [21]. Although small deviations in model parameters will not significantly affect GW interaction responses, an accurate Moon model is still of interest for precise restrictions on the GW of the source. Lunar seismometers, such as LGWA, could enhance models and locations with minimal seismic noise should be prioritized for optimal detection.

Further research could for example include studying gravitational waves coming from pulsars for Brans-Dicke polarizations. Future work should also consider the rotational effects of the Moon on eigenfrequencies, which has not been taken into account in this report.

Bibliography

[1] M. Kachelriess and M. P. Nødtvedt. "Lunar response to gravitational waves". In: (Dec. 2023). DOI: 10.48550/ARXIV.2312.11665. arXiv: 2312.11665 [gr-qc].

[2] Albert Einstein. "Näherungsweise Integration der Feldgleichungen der Gravitation". von A. Einstein. In: (1916).

[3] A. Einstein. "Die Feldgleichungen der Gravitation." German. In: *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften* 1915 (1915), pp. 844–847.

[4] Sean M. Carroll. *Spacetime and Geometry. An Introduction to General Relativity*. Cambridge University Press, 2019. DOI: 10.1017/9781108770385.

[5] The LIGO Scientific Collaboration and The Virgo Collaboration. "A Search for Tensor, Vector, and Scalar Polarizations in the Stochastic Gravitational-Wave Background". In: *Phys. Rev. Lett. 120, 201102 (2018)* 120.20 (Feb. 27, 2018), p. 201102. ISSN: 1079-7114. DOI: 10.1103/physrevlett.120.201102. eprint: 1802.10194.

[6] Hatice Özer and Özgür Delice. "Gravitational waves in Brans–Dicke theory with a cosmological constant". In: *The European Physical Journal C* 81.4 (Apr. 2021). ISSN: 1434-6052. DOI: 10.1140/epjc/s10052-021-09123-7.

[7] E. Poisson and C.M. Will. *Gravity: Newtonian, Post-Newtonian, Relativistic*. Cambridge University Press, 2014. ISBN: 9781107032866. URL: https://books.google.ch/books?id=PZ5cAwAAQBAJ.

[8] B.P. et al. (LIGO Scientific Collaboration Abbott and Virgo Collaboration). "Observation of Gravitational Waves from a Binary Black Hole Merger". In: *Physical Review Letters* 116.6 (Feb. 2016), p. 061102. ISSN: 1079-7114. DOI: 10.1103/physrevlett.116.061102.

[9] J.W. Rudnicki. *Fundamentals of Continuum Mechanics*. Wiley Desktop Editions Series. Wiley, 2014. ISBN: 9781118479919. URL: https://books.google.ch/books?id=x5zmBQAAQBAJ.

[10] A. Ben-Menahem. "Excitation of the Earth's eigenvibrations by gravitational radiation from astrophysical sources". In: *Il Nuovo Cimento C* 6.1 (Jan. 1983), pp. 49–71. ISSN: 0390-5551. DOI: 10.1007/bf02511372.

[11] W. W. Hansen. "A New Type of Expansion in Radiation Problems". In: *Physical Review* 47.2 (), pp. 139–143. ISSN: 0031-899X. DOI: 10.1103/physrev.47.139.

[12] T. B. A. Senior. "A NOTE ON HANSEN'S VECTOR WAVE FUNCTIONS". In: *Canadian Journal of Physics* 38.12 (1960), pp. 1702–1705. ISSN: 0008-4204. DOI: 10.1139/p60-173.

[13] Jose Pujol. *Elastic wave propagation and generation in seismology*. Jose Pujol. Cambridge: Cambridge University Press, 2003. 444 pp. ISBN: 9780511610127.

[14] A. Ben-Menahem. *Seismic Waves and Sources*. Ed. by S. J. Singh. Description based on publisher supplied metadata and other sources. New York, NY: Springer New York, 1981. 11127 pp. ISBN: 9781461258568.

[15] Raphael F. Garcia et al. "Lunar Seismology: An Update on Interior Structure Models". In: *Space Science Reviews* 215.8 (2019). ISSN: 0038-6308. DOI: 10.1007/s11214-019-0613-y.

[16] Josipa Majstorović, Séverine Rosat, and Yves Rogister. "Earth's spheroidal motion induced by a gravitational wave in flat spacetime". In: *Physical Review D* 100.4 (). ISSN: 2470-0010. DOI: 10.1103/physrevd.100.044048.

[17] J. H. Woodhouse and A. Deuss. *Theory and Observations – Earth's Free Oscillations*. 2007. DOI: 10.1016/b978-044452748-6.00002-x.

[18] Freeman J. Dyson. "Seismic Response of the Earth to a Gravitational Wave in the 1-Hz Band". In: *The Astrophysical Journal* 156 (May 1969), p. 529. ISSN: 1538-4357. DOI: 10.1086/149986.

[19] Yosio Nakamura and Junji Koyama. "Seismic Q of the lunar upper mantle". In: *Journal of Geophysical Research: Solid Earth* 87.B6 (June 1982), pp. 4855–4861. ISSN: 0148-0227. DOI: 10.1029/jb087ib06p04855.

[20] Jan et. al. Harms. "Lunar Gravitational-wave Antenna". In: *The Astrophysical Journal* 910.1 (Mar. 2021), p. 1. ISSN: 1538-4357. DOI: 10.3847/1538-4357/abe5a7.

[21] Ajith et al. "The Lunar Gravitational-wave Antenna: Mission Studies and Science Case". In: (Apr. 2024). DOI: 10.48550/ARXIV.2404.09181. arXiv: 2404.09181 [gr-qc].

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Appendices

A

Appendix A: Linearization of Cauchy's equation of motion

This appendix aims to simplify the differential equation derived from Cauchy's equation of motion in Chapter 2 (equation 2.19) under our assumptions.
We begin by exploiting these assumptions to manipulate the first term in Eq. 2.19. First, we can separate this term into an initial stress $\sigma^{(0)}_{jk,j}$ and an additional stress $\delta\sigma_{jk,j}$ after displacement. Using $(f\delta_{jk})_{,j}$ and our assumptions about initial stress in equation 2.30, we obtain a simpler form for the divergence of the initial stress tensor:

$$\sigma^{(0)}_{jk,j} = \left(-\left(p^{(0)} + g^{(0)}\rho^{(0)}u_r\right)\delta_{jk}\right)_{,j} = -\left(p^{(0)} + g^{(0)}\rho^{(0)}u_r\right)_{,k}. \tag{A.1}$$

Here, u_r denotes the radial component of displacement, not a vector.
Using assumption 2.31), we can now express the divergence of the additional stress in spherical coordinates as

$$(\delta\sigma_{jk})_{,j} = \left(\lambda u_{j,j}\right)_{,k} + \mu\left(u_{k,jj} + \left(u_{j,j}\right)_{,k}\right) + \frac{d\mu}{dr}\left(2\frac{\partial u_k}{\partial r} + \epsilon_{kab}\hat{e}^{(r)}_a\left(\epsilon_{bcd}u_{d,c}\right)\right). \tag{A.2}$$

Having simplified the first term of the equation, we proceed to the second term, which involves the density. The continuity equation allows us to express the density as a function of the initial density and displacement:

$$\rho(\mathbf{r}) = \rho^{(0)}(\mathbf{r}-\mathbf{u})\cdot(1-u_{i,i}) = \rho^{(0)}(\mathbf{r}) \underbrace{-\left(\rho^{(0)}u_i\right)_{,i}}_{\rho' := -\left(\rho^{(0)}u_i\right)_{,i}} \tag{A.3}$$

With the assumptions regarding the gravitational potential and a quick calculation, we can then write this second term as

$$\begin{aligned}\rho F_j &= \left(\rho^{(0)} - \frac{d\rho^{(0)}}{dr}u_r - \rho^{(0)}u_{k,k}\right)\left(\psi_j - g^{(0)}\hat{e}_j^{(r)}\right) \\ &= \rho^{(0)}\psi_j + g^{(0)}\left(\frac{d\rho^{(0)}}{dr}u_r + \rho^{(0)}u_{k,k} - \rho^{(0)}\right)\hat{e}_j^{(r)}\end{aligned} \tag{A.4}$$

Finally, by combining the simplified versions of each term in equations A.1 A.2 and A.4, we arrive at the desired differential equation under our specified assumptions:

$$\begin{aligned}&\mu\left(u_{k,jj} + \left(u_{j,j}\right)_{,k}\right) + \frac{d\mu}{dr}\left(2\frac{\partial u_k}{\partial r} + \epsilon_{kab}\hat{e}_a^{(r)}\left(\epsilon_{bod}u_{d,c}\right)\right) + \left(\lambda u_{i,i}\right)_{,k} \\ &+ \rho_0\left(\left(\psi - g^{(0)}u_r\right)_{,k} + g_0\hat{e}_k^r u_{i,i}\right) + \omega^2\rho^{(0)}u_k = 0.\end{aligned} \tag{A.5}$$

We now focus on the behavior of the gravitational potential itself. The potential satisfies the Poisson equation:

$$\Psi_{,jj} = -4\pi G\rho \tag{A.6}$$

By inserting our expansion of the potential $\Psi = \Psi^{(0)} + \psi$ into the Poisson equation, we find that the perturbation of the gravitational potential satisfies:

$$\psi_{,ii} = -4\pi G\rho' = 4\pi G\left(\rho^{(0)}u_j\right)_{,j}. \tag{A.7}$$

B

Appendix B: Wigner Symbols

This appendix provides a brief introduction on Wigner symbols. Clebsch-Gordan coefficients are mathematical objects used mostly in the context of angular momentum in quantum mechanics. They are defined to express the addition of two angular momenta in terms of a third:

$$|j_1, j_2; j, m\rangle = \sum_{m_1, m_2} \langle j_1, j_2; m_1, m_2 | j_1, j_2; j, m\rangle |j_1, j_2; m_1, m_2\rangle \tag{B.1}$$

respectively

$$|j_1, j_2; m_1, m_2\rangle = \sum_{j,m} \langle j_1, j_2; m_1, m_2 | j_1, j_2; j, m\rangle |j_1, j_2; j, m\rangle. \tag{B.2}$$

These expansion coefficients $\langle j_1, j_2; m_1, m_2 | j_1, j_2; j, m\rangle$ are called the Clebsch-Gordan coefficients. In some cases, the Clebsch-Gordan coefficients are written in terms of Wigner (3-j) symbols. We have

$$\langle j_1, j_2; m_1, m_2 | j_1, j_2; j, m\rangle = (-1)^{j_1 - j_2 + m} \sqrt{2l+1} \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix} \tag{B.3}$$

A first, and most important property for our thesis, is the following:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = 0 \quad \text{unless} \begin{cases} m_1 + m_2 + m_3 = 0 \text{ and} \\ |j_1 - j_2| \leq j_3 \leq |j_1 + j_2| \end{cases} \tag{B.4}$$

Furthermore, changing the sign of the m „quantum numbers“ gives a phase factor of $(-1)^{j_1+j_2+j_3}$:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} \tag{B.5}$$

This implies directly

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ 0 & 0 & 0 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ 0 & 0 & 0 \end{pmatrix} \tag{B.6}$$

or

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ 0 & 0 & 0 \end{pmatrix} = 0 \quad \text{if } j_1 + j_2 + j_3 \text{ is odd.} \tag{B.7}$$

We use the Wigner symbols primarily to do expand the spherical vector functions $C_i^{(ml)}$, $P_i^{(ml)}$ and $B_i^{(ml)}$ in terms of Wigner symbols. For this, we make use of the following property, which relates the spherical harmonics Y_l^m to the Wigner symbols:

$$\int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta Y_{l_1}^{m_1} Y_{l_2}^{m_2} Y_{l_3}^{m_3} = 4\pi \begin{pmatrix} l_1 & l_2 & l_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \tag{B.8}$$